

We have

$$\prod_{k=1}^n \left(1 + \frac{1}{k}\right) = n + 1$$

Proof

BASE CASE:

Let $n = 1$

$$\begin{aligned} \prod_{k=1}^n \left(1 + \frac{1}{k}\right) &= \prod_{k=1}^1 \left(1 + \frac{1}{k}\right) = 1 + \frac{1}{1} \\ &= 1 + 1 \\ &= \boxed{2} \end{aligned}$$

$$n + 1 = 1 + 1 = \boxed{2}$$

$$2 = 2$$

$\therefore n = 1$ holds true

INDUCTIVE CASE:

We assume

$$\prod_{k=1}^n \left(1 + \frac{1}{k}\right) = n + 1$$

We prove:

$$\begin{aligned} \prod_{k=1}^{n+1} \left(1 + \frac{1}{k}\right) &= (n + 1) + 1 \\ &= n + 2 \end{aligned}$$

$$\begin{aligned} \prod_{k=1}^{n+1} \left(1 + \frac{1}{k}\right) &= \prod_{k=1}^n \left(1 + \frac{1}{k}\right) \cdot \left(1 + \frac{1}{n+1}\right) \\ &= (n + 1) \cdot \left(1 + \frac{1}{n+1}\right) \\ &= (n + 1) \cdot \frac{n+1}{n+1} \\ &= (n + 1) + 1 \\ &= \boxed{n + 2} \end{aligned}$$

$$n + 2 = n + 2✓$$